Enterprise Cloud Data Platform Architecture & Specification v4.2

Confidential Technical Documentation & Runbook — Core Infrastructure Engineering

Section 1: Microservices Architecture & System Overview

The Enterprise Cloud Data Platform operates on an air-gapped, zero-cloud dependency architecture. The system decouples data ingestion, semantic vector indexing, hybrid retrieval, and generative inference into independent microservices. The ingestion subsystem utilizes IBM Docling for native layout-aware parsing of technical documents, preserving headers, tabular metadata, and code structures. Documents are segmented using recursive header chunking with a target window of 1500 characters and a 150-character overlap.

The storage layer utilizes an embedded Qdrant vector database operating in local disk persistence mode. To achieve enterprise-grade recall, Qdrant combines dense vector embeddings generated by BAAI/bge-small-en-v1.5 (384 dimensions) with sparse BM25 token frequencies, fused via Reciprocal Rank Fusion (RRF).

Section 2: High Availability, SLA Metrics & Disaster Recovery

The platform guarantees high availability and strict service level agreements (SLAs) across all tier-1 production regions. The defined SLA commitments and recovery objectives are detailed in the operational matrix below:

Service Metric	Target Threshold	Measurement Interval	Penalties / Breach Actions
Platform Uptime SLA	99.995% Uptime	Monthly Rolling	Tier-1 Credit Reallocation
Recovery Time Objective (RTO)	Under 15 Minutes	Per Incident Event	Automatic Standby Cluster Failover
Recovery Point Objective (RPO)	0 Seconds (RPO=0)	Continuous Sync	Zero Data Loss Synchronous Replication
P99 Query Latency SLA	Under 120 ms	Sub-second 15,000 req/s	Horizontal Pod Autoscaler Trigger
Cross-Encoder Rerank Latency	Under 85 ms	Top-15 Candidates	Thread Pool Concurrency Scaling

Section 3: Error Code Catalog & Incident Handling Runbook

Production systems must handle and log standard enterprise platform errors according to the following specifications:

ERR-409-TIMEOUT: Distributed lock acquisition timed out after 30,000 ms. This error is triggered when an active transaction in the distributed worker cluster fails to acquire or renew a mutex in the distributed lock coordinator due to partition lag or high contention on shared tenant state. Incident Resolution: 1. Terminate the stale transaction thread immediately to prevent cascading deadlocks. 2. Inspect Redis cluster shard health and active lock keys using CLUSTER NODES. 3. Trigger an idempotent rollback of uncommitted operations. 4. Client applications MUST implement exponential backoff with randomized jitter between 200 ms and 5,000 ms before retying.

ERR-502-BAD-GATEWAY: Ingress gateway upstream connection failure. Occurs when Envoy or NGINX ingress router times out after 5,000 ms attempting to establish a TCP handshake with the upstream Auth-Worker pod. Resolution: Check Kubernetes pod readiness probes and inspect cluster coredns resolver logs.

ERR-401-TOKEN-EXPIRED: Bearer authentication token has exceeded its maximum validity window of 3,600 seconds. Clients must exchange their cryptographically signed refresh token at /api/v1/auth/refresh to receive a new JSON Web Token.

ERR-429-RATE-LIMIT: Tenant quota exceeded. The standard rate limit is 5,000 requests per minute per authenticated tenant organization. Surplus requests are queued in the token bucket or dropped with a Retry-After response header.

Section 4: Security Compliance, Encryption & Zero-Trust Governance

The platform complies with SOC2 Type II, ISO/IEC 27001, and HIPAA compliance frameworks for healthcare and financial enterprise workloads. All data at rest stored within the embedded Qdrant vector database and attached block volumes is encrypted using AES-256-GCM. Cryptographic keys are managed via AWS KMS Customer Managed Keys (CMK) or HashiCorp Vault with automated 90-day key rotation.

All internal network communications enforce mutual TLS (mTLS) with TLS 1.3 protocol standards, eliminating unencrypted plain-text transit. Role-Based Access Control (RBAC) and attribute-based security policies restrict data access at the document chunk level. Auditing logs are immutably streamed to an append-only WORM (Write Once Read Many) compliant storage bucket.

Section 5: Cross-Encoder Reranking & Abstention Guardrail Specifications

To eliminate LLM hallucinations and optimize context precision, the RAG engine implements a two-stage retrieval pipeline: Stage 1 performs hybrid retrieval to fetch top-15 candidate chunks using Reciprocal Rank Fusion. Stage 2 routes these candidates through a local BAAI/bge-reranker-base cross-encoder model. Raw cross-encoder output logits are normalized using a sigmoid transformation: S_norm = 1 / (1 + exp(-logit)). An Abstention Guardrail is configured with a strict confidence threshold of 0.35. If the top-ranked chunk exhibits a normalized rerank score below 0.35, the pipeline immediately abstains from generation, returning is_grounded=False and the message 'Insufficient context in the knowledge base to answer this question reliably.'